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The Development of Implicit Leadership Theories During Childhood: A Reconceptualization Through the Lens of Overlapping Waves Theory

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Implicit leadership theories (ILTs) are people's lay theories, definitions, or conceptualizations of leadership. In adults, they determine what actions we perceive as leadership, influence to whom we grant leadership status, and shape our own behaviors when we want to be seen as leader. Naturally, there has been an enduring interest in how these ILTs develop in children. Current theorizing on the development of leadership conceptualizations in children aligns with a stepwise progression mirroring Piaget's stage-based approach to cognitive development. However, contemporary approaches to cognitive development, such as Siegler's overlapping waves theory (OWT), acknowledge that children's development is linked to cognitive success and failure. This article integrates the findings from empirical studies into children's leadership conceptualizations and reinterprets them against OWT. This reinterpretation resolves findings that align poorly with a stepwise approach and demonstrates a strong fit with OWT. As such, children's leadership conceptualizations develop by generating and testing cognitive approaches—physical-spatiotemporal, functional, socioemotional, and humanitarian—and instead of progressing through these in order and according to age, they display variation and selection, that with experience and exposure, lay down selective combinations, which often engage multiple dimensions simultaneously. Consequently, the development of children's understanding of leaders is nonlinear, can be multidimensional, and is based on trial and error largely in response to their experiences. The article concludes with a discussion of the implications for future research and practice.

Keywords: implicit leadership theory, overlapping waves theory, child development, cognitive development, social cognition

As adults, individuals make decisions about whom to elect as leaders of countries, whom to appoint to executive positions in organizations, and whom to follow on social media. We make these decisions based on whom we think has been a leader or whom appears to display the traits and behaviors we associate with leadership. These decisions are expressions of our lay theories of leadership. In making these decisions, we process environment stimuli against our lay theories of leadership, also known as implicit leadership theories (ILTs), to determine whom we believe to be a potential leader and our response to them. Research has shown that these ILTs are mainly established during childhood (Ayman-Nolley & Ayman, 2005) and are quite stable in adulthood (Offermann & Coats, 2018). Hence, if we want to understand the reasons why people vote, appoint, or follow the people they do, it is important to understand how these ILTs form during childhood.

Despite a long history of published research into the nature of children's perceptions of leadership, progress has been hampered by a fragmented and disparate collection of work. This article's first contribution is to integrate this body of work. Although scholars have been publishing work on children's perceptions of leadership for almost 100 years, for example, Broich (1929) to Stavans and Diesendruck (2021), a consistent language and

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theoretical framework for describing and examining ILTs is more recent with the majority of this research published in the last two decades. The second contribution this article makes is to reexamine the literature on children's perceptions of leadership through an ILT lens. While ILT has become the dominant lens to examine the lay theories of leadership held by working adults, overlapping waves theory (OWT) has emerged as primary contemporary theory to explain children's cognitive development (Siegler, 1996, 2000). Thus, our third contribution is to integrate these two approaches to describe and explore the development of ILTs during childhood more richly. In this article, we define childhood broadly as young people of preschool, primary school, and secondary school age, noting that much of the relevant literature focuses on primary school age children.

The structure of this article first provides an introduction to ILTs and their role in social cognition. We then review prior scholarly work into the way children develop their understanding of leadership. Next, we explain OWT and compare and contrast it with the previous theories of cognitive development during childhood. Following the discussion of the focal theoretical frameworks, we conduct a scoping review of the literature related to the development of children's understanding of leadership before applying an OWT lens to this body of literature.

Social Cognition and Implicit Leadership Theories

Social perception is a complex area of research within the social sciences. Its vast literature addresses, among many other issues, researchers' interest in understanding how people read and understand the actions of "the other" (Plaks et al., 2009). Specifically, cognitive information processing in humans has been found to rely on lay beliefs or traits that shape perception of "the other" and the environment and minimize mental work (Hong et al., 2001; Levy et al., 2006; Molden & Dweck, 2006). These cognitive processes, known as social cognition, include all information gathered from perceptive cues that are interpreted for emotional, behavioral, and interpersonal communication (Suchy & Holdnack, 2013). These processes determine everyday working frameworks for social interaction and the phenomenon of leadership, which is social in its core, is subject to these processes of social cognition.

Stogdill (1948) summarized the literature of features describing leaders and structured the research on traits into personal factors including capacity, achievement, responsibility, participation, and status. He also acknowledged the relevance of the mental level, status, skills, and needs of followers and their role within the phenomenon of leadership (DeHaan, 1962; Stogdill, 1948). The observation that follower needs and evaluations of leader traits were relevant to leader emergence and subsequent evaluations of leader performance provided the embryonic foundation for a more formal investigation of schemas for leadership. However, it was not until 1975 that the study of traits was conceptually defined under the appellation of ILTs.

The ILT concept was formally introduced by Eden and Leviatan (1975) referring to beliefs held by individuals about how leaders behave in general and what is expected from them by others (Eden & Leviatan, 1975). They derived the concept from D. J. Schneider's (1973) implicit personality theories and as a response to developments in social cognitive theory. Since then, there has been a growing interest and increasing research activity in the study of significance of personality and traits for leadership (Felfe & Schyns, 2014), and ILTs have come to occupy an important place in the leadership literature (Lord et al., 2020).

ILTs are defined as lay images of leadership that individuals hold about the traits and behaviors of leaders (Eden & Leviatan, 1975; Epitropaki & Martin, 2004; Kenney et al., 1996; Offermann et al., 1994; Schyns & Meindl, 2005; Schyns & Schilling, 2011). In other words, ILTs are idiosyncratic conceptualizations (prototypes) that are employed as everyday images of what leaders are like (Foti et al., 2014; Lord & Maher, 1991; Lord & Shondrick, 2011; Offermann et al., 1994; Schyns & Schilling, 2011). For example, if people receive an email from a senior executive of one of the biggest companies in their country asking for a meeting, they will activate a set of cognitive presumptions of what this person's attributes will be like. Perhaps they imagine a well-groomed, intelligent, assertive, and dynamic male leader. The traits that come to mind are linked to their assumptions of what such leader is like; they have been contextually informed and developed by the exposure they have had to people in leadership positions throughout their lives and are an expression of their ILTs. These observer-centric images are important because they are used by perceivers to help them read and codify "the other" and respond correspondingly to leadership processes with minimal cognitive effort (Billsberry et al., 2018; Shondrick et al., 2010). ILTs ignite the beginning of the leadership process and provide the groundwork for the development of leader–follower interactions. Perceivers use ILTs in a dynamic and integrative way that employs the use of mental categories and schemas "that affect both perception and memory" (Lord et al., 2020, p. 51) to make rapid sense of another's intentions and behaviors (Shondrick et al., 2010). Consequently, ILTs are a mental model of leadership from the observer's point of view (Billsberry & Meisel, 2009) and are thought to be central in the leadership process because they determine the leader's expected attributes, roles, and privileges (Lord & Maher, 1991). Lord et al. (2001) suggested that a cognitive matching process is triggered when context stimuli relevant to leadership are encountered. These are automatically and instantaneously matched to people's ILTs to determine whether or not leadership has been witnessed. ILTs also determine which leaders will be more likely to be accepted and allowed to exert influence (House et al., 2002; Junker & Van Dick, 2014; Kenney et al., 1996). Once sensory stimuli from observations of a person are perceived, compared, and subsequently matched to someone's ILT, the individual is labeled a leader and other related constructs such as the ability to influence others are also activated (Lord & Shondrick, 2011).

As a result, ILTs are influential in the expectations that followers bring to the leader–follower relationship (Offermann et al., 1994). So, the acceptance of someone as a leader is only possible if there is a match between potential followers' prototypical mental model of the leader and their tangible perception of that person (Billsberry et al., 2018; Van Quaquebeke & Van Knippenberg, 2012). ILTs help guide initial perceptions and the active construal of others by providing a set of generic assumptions and beliefs as what to expect and how to respond in an adaptive manner to various individuals (Brown et al., 2004; Fiske & Taylor, 1984; Shondrick et al., 2010). When ILTs match situational perception, the process of claiming and granting leader distinctiveness is ignited (DeRue & Ashford, 2010; Schyns et al., 2011) and followership behavior emerges (Swanson et al., 2020).

The structure of cognitions in ILTs have been observed through knowledge structure theory, across multiple types of information-processing models in leadership perceptions and ILT development (e.g., symbolic views, connectionist views, and embodied embedded views; Lord & Shondrick, 2011). For example, categorization theory states that people gradually build general knowledge about leadership, developing fixed prototypic ideas shaped by the perception of features that transform information into categories (Engle & Lord, 1997; Lord & Maher, 1991; Lord et al., 2001) or independent mental exemplars (Swanson et al., 2020). Connectionist models proposes that knowledge can be stored across neuronallike processing units, retrieved unconsciously, and processed very quickly (Lord & Shondrick, 2011). These parallel units stay dormant but once stimulated, they activate, creating patterns, or "neural nets" that have different weights (Lakomski, 2001). As such, connectionist systems can dynamically adjust category prototypes to fit contexts (Foti et al., 2017), providing a combinatory platform where the individual, the group, and the context can interconnect (Lord et al., 2001). Embedded leadership impressions are formed initially by the immersion of an individual within an environment, which frames cognition prior to accessing cognitive knowledge mandated by the previous experience or conceptual categories and schemas as those proposed by the connectionist and categorization theories. In this view, the structures of cognition are seen as either emotional, perceptual, or motor reactions to leaders which are embedded in the network of patterns that make a person's leadership knowledge. These patterns are accessed for sensemaking processes and influence follower behavior (Naidoo et al., 2010).

Factors That Influence ILT Variation

ILTs vary because every person has distinctive exposure to, and experiences with, leaders (Shondrick et al., 2010) and because individuals are immersed in diverse demographic, cultural, educational, and interest-based groups. As a result, numerous factors have been found to impact or mediate the perceptions of leadership and behavior. ILTs are shaped by an inward process linked to the perceivers' self-concept (Catrambone et al., 1996; Engle & Lord, 1997; Fong & Markus, 1982; Offermann et al., 1994) through an ongoing comparative process by which individuals simultaneously process information and search for similarity of characteristics and behavior in "the other" (D. Byrne, 1971; Dulebohn et al., 2017; Engle & Lord, 1997). Consequently, people follow leaders that are similar to themselves demographically, culturally, and ethically (D. Byrne, 1971; Dulebohn et al., 2017; Engle & Lord, 1997; Greenwald & Banaji, 1995). In addition, factors such as early interactions (Bass, 1990; Brungardt et al., 1997), gender (N. Cantor & Mischel, 1979; Offermann et al., 1994), culture (Gerstner & Day, 1994; Koopman et al., 1999; Offermann et al., 1994), race (Rosette et al., 2008), power (Konst & Van Breukelen, 2005; Palich & Hom, 1992), hierarchical level (Shondrick et al., 2010), and stereotypes (Kenney et al., 1996; Schyns & Schilling, 2011) have been found to impact ILTs. In addition, ILTs have been found to be influenced by memory (Epitropaki & Martin, 2004; Lord et al., 2020; Offermann et al., 1994) and language (Fairhurst & Grant, 2010; House et al., 2002; Kenney et al., 1996; Schyns et al., 2011; Shondrick et al., 2010).

Socially Determined ILTs

Even though ILTs are held individually and ignited through the perception of "the other" based on the perceiver's implicit ideas of what leaders are (Den Hartog et al., 1999; Lord et al., 1984; Lord & Maher, 1991), ILTs can also be collective and socially determined (van Knippenberg et al., 2004). This means that in a group, even if the potential leader matches one person ILTs, leadership status will not be achieved if the other group's members do not hold comparable ILTs (Eden & Leviatan, 1975; Schyns et al., 2011). In other words, ILTs influence both one-on-one interactions between two people, as well as one-and-all interactions between a potential leader and a group of people.

Since leadership stands within the spectrum of behaviors within groups of people, it explains why groups often possess multiple, contextually based schemas and categories of leaders (Lord et al., 1982, 2001; Shondrick et al., 2010). For example, Lord et al.'s (1984) study of leader content prototypes found that the group of study, even when rating different types of leaders, considered "intelligent," "outgoing," "understanding," and "dedicated" as leadership attributes (Lord et al., 2020). In other words, people's ILTs work congruently alongside each other, responding to the variability of contexts and nature of social groups creating patterns of features, attributes, and behaviors which result in leadership syntality, a collective mental image of leadership (Shondrick et al., 2010). So, only when the ILTs of leaders match sufficiently across patterns in a group, or group leadership structures (Lord et al., 2020), are leaders granted leader identity, the relationships become clear, and leaders are collectively recognized (DeRue & Ashford, 2010). Hence, the concept of ILTs is complex as not a unitary construct. On one hand, it is collectively molded and on the other hand, it works to unchain other constructs that can only appear once the ILTs are activated to help perceivers simplify social processes (Shondrick et al., 2010).

The Development of ILTs During Childhood

All this evidence explains why ILTs have come to occupy an important place in the leadership literature (Lord et al., 2020). While most of the body of literature focuses on understanding how these lay theories manifest in adults and in workplace environments, there is very limited research exploring how these lay theories manifest in earlier stages of the human development, such as the formative childhood years. This emphasis, since the 1970's, on adult ILTs has provided rich insights into how individuals *apply* ILTs (Ayman-Nolley & Ayman, 2005; Lord et al., 2020). However, far less is known about how individuals *form and develop* ILTs.

In terms of ILT development, we know that ILTs form in early childhood (Antonakis & Dalgas, 2009; Ayman-Nolley & Ayman, 2005; Borman, 1987; Oliveira, 2016; Stavans & Diesendruck, 2021) and develop throughout the school years (Frost, 2016; Shondrick et al., 2010). And that formative periods in childhood are important for adult leadership behaviors because adults tend to rework childhood leadership scenarios in the workplace (Keller, 1999) and managerial styles have been found to relate to chief executive officers' life experiences such as childhood relationships (Bernile et al., 2017). We also know that early childhood experiences have an impact upon individual differences and influence variations in subsequent adult ILTs (Hunt et al., 1990; Keller, 1999; Ligon et al., 2008). For example, Ligon et al. (2008) found that significant life events and their linked narratives appear to shape mental models of behavior in outstanding leaders. All this evidence inevitably raises questions about the extent to which our adult ILTs are formed during childhood.

The next step toward understanding how ILTs emerge and develop is to appreciate that there have been many years of scholarly work exploring the way that children's conceptualizations of leadership develop before the nomenclature of ILTs emerged. Although these studies do not use the language of ILTs, they explore the same phenomenon and thus one of this article's contributions is to integrate that literature with studies from more recent times. However, before we discuss the development of leadership conceptualizations in children, we first review theories of cognitive development in childhood, as they serve as the underlying mechanisms upon which leadership conceptualizations emerge.

Cognitive Development During Childhood

Studies of the development of children's conceptualizations of leadership are underpinned by broader theories of cognitive development during childhood. Most of the studies exploring children's conceptualizations of leadership were conducted at a time when the ideas of Piaget and Vygotsky dominated the landscape of cognitive child development and such theories naturally influenced how researchers of children's ILTs interpreted their results.

Published in the 1930s, 1940s, and 1950s, Piaget's theories suggest children's cognitive development progresses toward logical thought, from concrete to abstract, across stages (Cohen, 2017; Robson, 2006). Piaget's progression is invariant, meaning that a child cannot move up a stage without being ready, and all children follow the same progression (Piaget, 1964). Since its inception, Piaget's theory has been challenged primarily because its sole focus is on logical thinking excluding other ways of reasoning (Bruner & Weinreich-Haste, 1987), because younger children show more advanced logical and abstract thinking than Piaget proposed (Donaldson, 1978; Gillett-Swan, 2017; Isaacs, 2015), and because it lacks reference to societal and emotional aspects (Thornton, 2002).

Although written and published in Russian in the 1920s and 1930s but only gaining notoriety in English in the 1970s, Vygotsky's post-Piagetian theories focused on social interactions as an avenue for enhanced learning and discovery (Cohen, 2017; Robson, 2006) where children's highest learning potential is dependent on collaboration with more experienced peers (Vygotsky, 1978). Vygotsky's ideas have been criticized for their focus on shared activity (Eun, 2019), the idea that not all experienced peer-to-one interactions result in advancement, and they can also be inhibiting and cause retrogression (Robson & Flannery Quinn, 2015; Siegler et al., 2003).

Siegler's discoveries in the 1980s triggered interest toward the study of variability, strategy, and change within domain-specific territories of children's thinking (Goldin-Meadow & Alibali, 2002; Hosenfeld et al., 1997; Shrager & Siegler, 1998). In particular, studies have shown that children use an array of strategies to solve a problem, whereas previous approaches (Piaget) suggested single strategies. These studies also found that change can be gradual and strategies can be layered. This discovery led to the development of Siegler's overlapping waves model which has become widely influential in child development theory (Cohen, 2017; Robson, 2006).

OWT (Siegler, 1996; Siegler et al., 2003) acknowledges that children's development is linked to both cognitive success and failure which produces elastic change. When challenged, children generate diverse strategies and by increasingly moving through trial and error, they test the usefulness of the strategies. As they become more experienced solving different and equivalent problems, preferences emerge for those strategic paths that prove more effective or more appropriate. In this sense, recurrent strategies survive and the more occasional abate. For example, when encountering a mathematical addition, a child can begin by using the counting the fingers strategy, which is a modeling out approach. As the task gets more complex, they introduce a second strategy, which can be counting up. As they gradually master early strategies they can also introduce new strategies, for instance, patterns in addition, or number grids. Eventually, they may begin to show preference for a strategy or a multistrategy approach, though they can always go back to an early perspective when needed.

OWT allows for the interlaying of cognitive dimensions that belong to specific strategies of thinking resulting in simultaneous modes of cognition, ongoing conceptual restructuring, and hierarchic integration and segregation (Robson, 2006; Siegler, 1996). According to Siegler's OWT of information processing (Siegler, 1996, 2000; Siegler et al., 2003), each child can be pictured swimming mentally within a wide "ocean" founded over cognitive dimensions. In a wavelike model, through trial and error in the experience of leadership, each child navigates transversely through cognitive dimensions creating superposed waves of thought over time and preferred mental pathways when processing leadership stimuli. For example, a child may appreciate, after varied cognitive trials, that the leader who is the biggest, may not be as effective or beneficial as the functional leader. Furthermore, since the model also allows for combinatory dimensional processing, in the same example, a child may turn toward a functional and sensitive leader, rather than settling within a one-dimensional perception. Siegler's model also implies that, once a particular wave starts, it continues, even though other waves may appear. And, because in this model, all waves influence each other and the overall tide, children's ILTs, for example, would become more sophisticated, becoming broader and more nuanced as they settle upon their understanding of leadership after successive episodes of cognitive trial and error. As children have different leadership experiences, each child's ILT is idiosyncratic, but this idiosyncrasy is bounded by societal forces around definitions of leadership.

This shift in developmental theory accommodates diverse thinking, variety in information-processing strategies, and the influence of experience rather than age. It has superseded phase approaches and opened the door for novel dynamic cognitive development models capable of capturing individual capacities for abstraction, differentiation, and elaboration on knowledge (Robson, 2006). Furthermore, it has surfaced further linkage between children's and adults' conceptual understanding. For example, brain research has showed that attentional systems in the brain are structurally equivalent in adults and children, only the younger ones are less informed, basically because these have been less used (Baars & Gage, 2010). Moreover, children can improve understanding and construct sophisticated mental models when they are familiar with and encouraged with experiences and practice, similar to adults (Meadows, 1993; Robson, 2006). Further, OWT allows precise attention to short intervals (Siegler, 2007), it has accurately depicted trends of variability and provided a model that shapes developmental change (Adolph et al., 2008).

Support for the model comes from commonalities found across studies on the development of mathematical thinking (Jansen & van der Maas, 2002; Shrager & Siegler, 1998; van der Ven et al., 2012), in nonalgorithmic domains such as spelling (Rittle-Johnson & Siegler, 1999), where data have shown that children utilize a minimum of three strategies, and in more qualitative domains (e.g., Hatano et al., 1993; Twissell, 2018; Vosniadou & Brewer, 1992). Focusing on the nonalgorithmic arena such as leadership, in their multinational study exploring children's understanding of the life concept, Hatano et al. (1993) found that children showed a greater understanding than would be expected through Piaget's stage-based approach. Same-age children from different countries presented differences in the animate versus nonanimate distinction, and these were influenced by linguistic categories and observations of the world including adults' behavior toward living things, everyday conversations, and exposure to educational content at school and in media. Vosniadou and Brewer's (1992) experiment investigating primary school children's conceptual knowledge about the earth found that the process of change across mental models is slow and gradual, which causes the emergence of interposed models. Such models arise through trial and error, as children reconcile the information that they receive on the matter with their initial mental models or presuppositions that the earth is rectangular. As they acquire knowledge, their explanatory frameworks change, and with ongoing theory restructuring, eventually they harmonize with the culturally accepted view that the earth is a sphere. Twissell (2018) explored conceptual representation and change in the development of electronics concepts. The author found four cognitive profiles emerging from the use of multiple representations when interpreting electronics functioning. These included approaches to how the students created understanding of electronics. One cognitive approach was based on observable physical features, others in symbolic notions, others functional, and the fourth presented a multidimensional understanding.

Taken together, these findings across a range of cognitive functions suggest that OWT is also appropriate to the development of ILTs and offers opportunities to enlighten our understanding of how these mental structures change and acquire cognitive growth.

Scoping Review

To find studies focusing on children's conceptualizations of leadership, we conducted a scoping review of the literature. Scoping reviews are appropriate when researchers are seeking a way into a literature (Grant & Booth, 2009). Specifically, our strategy was to find empirical studies that had examined children's leadership conceptualizations, where the children themselves reported on their own ideas and understanding of leadership. We wanted to exclude studies where adults (e.g., teachers and parents) commented on children's leadership conceptualizations (e.g., Cascio, 2016; Harrison et al., 1971; Stray, 1934; Terrell & Joy, 1958), as these observations are third-party assessments filtered through adults' conceptualizations (ILTs) of leadership. Once these studies were identified, we reviewed the findings related to the development of children's leadership conceptualizations and mapped these on to wave theory.

Although our preference was to adopt a Cochrane-style systematic methodology (Henderson et al., 2010), initial scoping work (in APA PsycArticles, APA PsycInfo, Scopus, and Web of Science) demonstrated that this method would be problematic for this literature. The problems thwarting the use of a traditional systematic approach included: (a) the broad range of terminology used to describe leadership conceptualizations (e.g., perceptions, mental models, implicit theories, cognitive frameworks, lay theories, and definitions); (b) the sheer volume of different words can be used to describe children (e.g., adolescent, boy, child, children, girl, infant, juvenile, kid, minor, pupil, school age, schoolboy, schoolgirl, student, teenager, young person, youngster, and youth) and the volume of different types and purposes of studies using these words made this search term impractical; (c) as the digitization of library resources weakens the further back in time ones goes, library-based searches lose effectiveness when an extended period is explored (Jonkers & Derrick, 2012; W. Liu, 2021); and (d) searches were further complicated by relevant articles being published in languages other than English.

Acknowledging these challenges, we decided to employ a multistage process involving backward snowballing (Badampudi et al., 2015). Through a systematic scoping review (Grant & Booth, 2009), we wanted to identify a core of relevant contemporary articles on the topic. These would be the starting point for qualitative citation analysis. In other words, we wanted to find a corps of relevant articles whose citations we would search to find other articles on the subject. Once new articles were identified, we would search the citations on those to find further articles. This backward snowballing ended when each avenue of investigation ran dry.

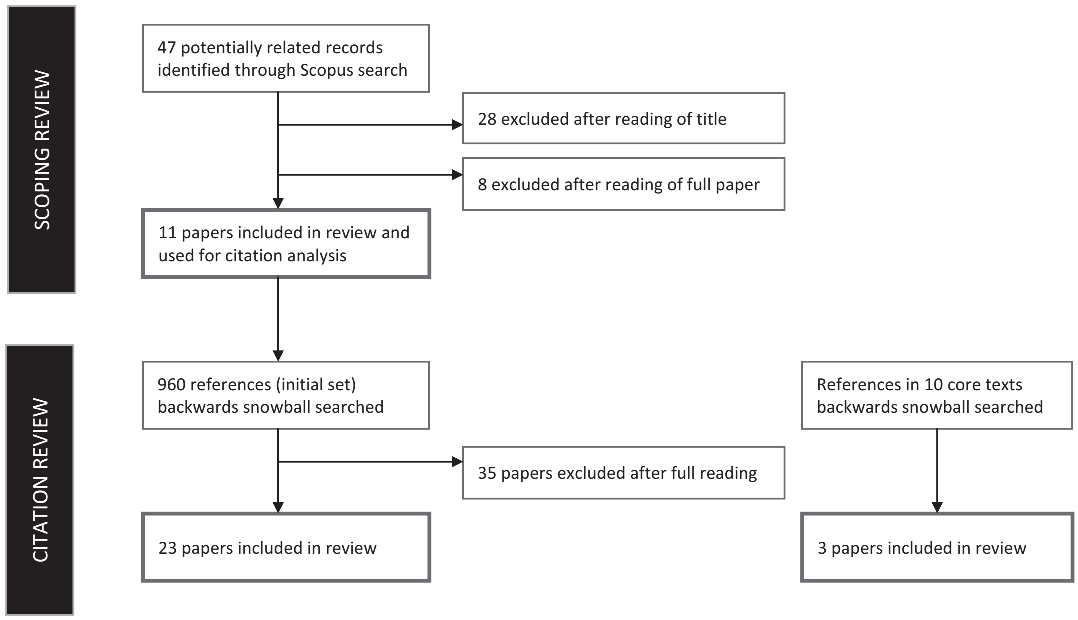
The scoping element of the search was conducted in Scopus with the following syntax: (TITLE-ABS-KEY ({children} OR {youth} OR {infant} OR {school-age} OR {primary school} OR {elementary school} OR {child}) AND ALL ({implicit leadership theory} *OR* {implicit leadership theories})). This syntax searched for articles that contained children-related terminology in the title, abstract, or keywords, and ILT anywhere in the articles. We chose Scopus for the scoping as it is known to offer strong coverage of the leadership literature and has 80%–90% overlap with other databases such as Web of Science and Google Scholar (Martín-Martín et al., 2021).

As illustrated in Figure 1, the scoping review surfaced 47 candidate articles of which 11 contained reports of empirical studies, where the children themselves reported on the nature of their leadership conceptualizations. Of the 11 articles, six were framed as ILT studies and five were not. One article, Ayman-Nolley and Ayman (2005), reported four separate studies.

The articles referenced by these 11 studies were searched for further studies also looking into children's conceptualizations of leadership. Through the backward snowballing approach, when an article was discovered, the references of that article were further examined for more articles. The process continued until each line of tracing ran out. This process surfaced 58 candidate articles of which 23 were included in the review after the articles were read. In addition, we also conducted a backward snowballing citation search starting in some key leadership texts (e.g., Eden & Leviatan, 1975; D. J. Schneider, 1973; Stogdill, 1948). This yielded three more candidate articles, all of which met the criteria for inclusion. In total, the database for this scoping review comprises 40 studies in 37 articles.

The authors read these articles taking copious notes relating to method, sample, findings, and implications. In writing the various sections of the article, the authors integrated and synthesized findings from across the various studies aligning them with OWT. Where there were disagreements, these were discussed.

Figure 1
Flow Diagram of Literature Search and Filtering



A summary of the articles included in the review can be found in Tables 1 and 2. The first table contains eight articles conducting investigations under the umbrella of ILTs; the second contains 29 articles exploring children's conceptualizations of leadership outside this perspective. Table 3 aligns empirical findings with OWT.

All the data gathered for this scoping review are in the public domain as they are published journal articles. These articles are listed in Tables 1 and 2, and the details for each one is in the list of references at the end of the article. The syntax for our initial search has been detailed above. This study was not preregistered.

Overlapping Waves Theory Applied to the Development Implicit Leadership Theories During Childhood

Consistently, research shows that children in kindergarten and as young as five hold a concept of a leader (Ayman-Nolley & Ayman, 2005; DeHaan, 1962; Lord & Maher, 1991) and can often distinguish between leaders and nonleaders (Antonakis & Dalgas, 2009). Once leadership schemata are ignited in cognition, children's conceptualizations of leadership develop as they age, as they become more socially aware, and as they experience or witness leadership (Ayman-Nolley & Ayman, 2005; Broich, 1929; Chauvin & Karnes, 1984; DeHaan, 1962; Frost, 2016; L. Liu et al., 2012; Oliveira, 2016; Pigors, 1933; Sacks, 2009; Salmond & Fleshman, 2010; Selman et al., 1977). This development has shown that from 5 years onward, children use four approaches when trying to make sense of leadership (see Table 3): physical-spatiotemporal, functional, socioemotional, and humanitarian (Broich, 1929; DeHaan, 1962; Selman et al., 1977).

Physical-Spatiotemporal Approach

The physical-spatiotemporal approach is where observable stimuli about the leader are processed. In this cognitive dimension, children navigate through physical and spatial understandings of leadership linked to observations of leadership roles (e.g., the one in front; Broich, 1929; Selman et al., 1977), social interactions based on authority (Selman & Jaquette, 1977), and to specific role models (Lord & Maher, 1991). This is probably because within this dimension children hold a definition of leader based on their response to stimuli provoked by their observable level of perception (Lord & Maher, 1991). The core social functions of this wave are to help perceivers' primary knowledge of a person's potential for leadership through noticeable observable cues (physical or spatial) and first-hand experiences of the provision of leadership status to "the other" (DeHaan, 1962).

Physical cues have been found to be central to other developmental theories. For example, social dominance awareness theory has found that morphological traits serve as social cues when decoding dominance or nondominance in children as young as 4 years old (Keating & Bai, 1986). Similarly, spatial cues on social dominance awareness have been found in toddlers, as they begin to move autonomously (Mascaro & Csibra, 2012), where they also use predictions based on spatiotemporal parameters, including size, for dominance outcomes (Thomsen et al., 2011). Features such as nonsmiling faces, greater facial width, and larger jaws are associated with social dominance across cultures (Keating & Bai, 1986).

Functional Approach

The functional approach is where task-based stimuli and the leader's functionality are processed. Through this cognitive

Table 1
Empirical Studies of Children's ILTs

Author	Sample size	Age/grade	Context	Methodology	Main finding
Matthews et al. (1989) cited in Lord and Maher (1991)	160	Grades 1, 3, 6, 9, and 12	United States	Interviews	Explored development of leadership categories. Younger children presented less sophisticated categories of leaders, bottom-up definitions, and referred to exemplars more often than prototype models of categorization. Older children show more abstract definitions. Noted that 5-year-old children can differentiate between a leader and a nonleader, they do not render high levels of importance to such divisions, and children between 12 and 18 years old give more importance to the leader/nonleader division. The authors noted that cumulative experience with leaders develops the capacity for abstract cognitive thinking of leadership.
Ayman-Nolley and Ayman (2005), Study 1 (1989)[a]	36	Grade 3	United States	Drama play writing and production followed by interview	Children in Grade 3 have ILTs and may already hold a gender schema of the leader as male. Most children expect leaders to have followers. Children's ILTs can be positive or negative, related mostly to task accomplishment, but also to being active and concerned for others.
Ayman-Nolley and Ayman (2005), Study 2 (1992)[a]	471	Kindergarten to Grade 8	United States	Children's drawings. They compared children's characteristics (gender, ethnicity, and grade) with the drawing characteristics	Most children drew White male leaders in a peaceful environment with followers around. Some children also drew female leaders and Black leaders. Children overall identified 14 major role categories; the highest in frequency were person, entertainment, parade leader, and military. They found gender differences in leader perceptions, specifically on the role categories of the leader drawn (girls drew more teacher, child, and parent leaders, more smiling leaders; boys drew considerably more male leaders than girls, more sports and military leaders, and have more tendency to draw violent pictures). Most of the violent referenced leaders were drawn by White boys. The authors found that African American children drew more often leaders from their ethnicity. They found significant differences across grades in leader category (fewer categories in younger children), leader smile (more in younger children), and the presence of followers (highest in middle grade).
Ayman-Nolley and Ayman (2005), Study 3 (1993)[a]	130	Grade 3 and Grade 6	United States	Interview	Children showed a gender schema of a male leader that is stronger in the higher grade, and weaker in girls. Also, the leader's roles vary across gender and grade.
Ayman-Nolley and Ayman (2005), Study 4 (2003)[a]	352	Kindergarten to eighth grade (except seventh grade)	United States	Drawings and interpretable identification checkpoint	Children as young as kindergarten have ILTs. Eight-year-old children recognize the leadership–followership relationship and differentiate between good and bad leaders. Also, as children grow older tend to have a more skeptical view of leaders and children's leader category varies depending on their ethnicity. The authors found no relationship between children's gender, ethnicity, and grades. Children drew different physical and behavioral features to male leaders or female leaders.

(*table continues*)

Table 1 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
Ayman-Nolley et al. (2006)	Unknown	Unknown	Costa Rica and United States	Drawings	Results were consistent with the belief that ILTs vary in terms of gender and cultural background, though this research did not find conclusive data on the existence of key developmental trends of ILTs across grades or within genders. In terms of cultural background, the authors found that children drew mostly a male leader though there were differences between U.S. children and Costa Rican children ILTs. For example, U.S. girls drew more female leaders than Costa Rican girls, and leader roles varied as well; Costa Ricans drew more generic leaders and U.S. children drew more historical figures. Older children offered more complexity in their leader categories.
Sacks (2009)[a]	Sample 1: 42 Sample 2: 30 Sample 3: 164	Sample 1: Grades 5–8 Sample 2: Grades 9–12 Sample 3: Grades 4 to Grade 1 to Grade 8	Canada	Mixed methods. Focus groups (six—eight students each) and survey	The author found differences across grades in the way children define the leader's behavior, capacity, and the type of leadership they wish to undertake (from task-oriented to responsibility-oriented, to role-oriented and lastly, identity-oriented).
L. Liu et al. (2012)[a]	514	Grades 1–8	China	Drawings and two written short sentences about drawing	The authors found the prominence of a male leader in the children's drawings, because boys mostly draw male leaders. They also found that most children drew generic persons as leaders. Youngest children more often draw a generic person than the oldest who more often depict a leader with a social role. Gender of the leader's drawing was dependent on the kind of social role depicted. In addition, in comparison to the U.S. studies by Ayman-Nolley and Ayman (2005), Chinese girls more often drew male leaders than U.S. girls.
Oliveira (2016)[a]	Sample 1: 28 Sample 2: 175	8–11 years old	Philippines (Christian elementary schools)	Mixed methods combining children's drawings, interviews, and a questionnaire. Children were asked to "draw a picture of your favorite leader" and respond to questions regarding the drawing exploring their perceptions of gender, social role, and typology	Children's ILTs are highly sensitive to religious affiliation/education. Jesus was the most frequently cited leader by the children who also showed a preference for the male stereotype for leadership. Male prevalence for children's ideas of leader. The author found that children are likely to portray leadership models that are closer to their daily life, that stimulate their imagination or ideas, and that are valued in their culture as desirable. Differences found in age groups where the oldest children had a higher tendency to nominate a male leader and a religious one than younger children. Catholic girls were more likely to depict a female leader than Evangelical and Adventist girls. Children frequently nominated similar leader roles as children in China, United States, and Costa Rica (parents, teachers, and political figures), however, they differed by presented a high frequency of religious figures (Jesus).

(*table continues*)

Table 1 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
Frost (2016)[a]	Sample 1: 10 student leaders Sample 2: 30 student leaders, 69 student group members, 32 adults	16–19 years old (and also adults ~41 years old)	United States	Sample 1: Leadership statement cards sorting Sample 2: Questionnaire	Secondary school students have ILTs that are very similar to those held by working adults, which suggests that ILTs may be fully developed by early high school. Results suggest that experiences with leaders in groups influence the development of ILTs and also differences in leadership expectations between boys and girls.
Walker et al. (2020)[a]	102	Adolescents and adults from the Fullerton Longitudinal Study	Varied	Surveys to participants and parents during their adolescence and again at age 38	Adolescents raised in family environments high in conflict and focused on achievement are more likely to endorse tyranny as an ideal leadership characteristic more than 20 years later in adulthood. In contrast, adolescents raised in intellectually stimulating family environments are less likely to endorse tyranny as an ideal leadership characteristic, as are adolescents whose mothers engaged in effective limit-setting.

Note. ILT = implicit leadership theory.
[a] Surfaced in the scoping study.

dimension of information processing, children acknowledge the leader's functionality in terms of task and performance (Broich, 1929; DeHaan, 1962) and hold wider contextual input than when processing information within the physical-spatiotemporal dimension (Ayman-Nolley & Ayman, 2005; Broich, 1929; DeHaan, 1962; Pigors, 1933; Selman & Jaquette, 1977; Stogdill, 1948). Hence, this functional understanding is linked to realistic conceptions within a context (Broich, 1929; DeHaan, 1962; Pigors, 1933). In addition, they recognize the leadership–followership relationship (Ayman-Nolley & Ayman, 2005; Selman et al., 1977), where there are benefits and responsibilities in both leaders and followers (Selman & Jaquette, 1977). However, it is purposeful and rarely unfolds personality, social, or emotional traits of leaders. Information processing is capable of being discriminated between task and maintenance roles (DeHaan, 1962), and leader effectiveness can be measured within positive and negative notions of leadership (Ayman-Nolley & Ayman, 2005) and perceptions of effective leadership behavior (Yarrow & Campbell, 1963).

Consequently, when children process perceptive stimuli within this approach, they associate leadership with excellence, functionality, knowledgeability, and top-down authoritarian interactions (Nemerowicz & Rosi, 1997; Selman et al., 1977). However, because they focus on the leadership's purposefulness, in other words, what the leader's existence is for, what it can do, and how leadership can be used to achieve common and personal goals (DeHaan, 1962; Pigors, 1933), children's thinking within this action-based territory is open to granting leadership status to a peer, and not only to adults. Ways of thinking within this approach revolve around ideas of common cause (Pigors, 1933), mutuality and cooperation (Selman et al., 1977), the sociality of the game, and collaborative play (Piaget & Gabain, 1932). From a social perspective, in this approach, children develop an understanding of the relationship between an individual's conceptualization of a leader and the group's dynamic, including its objectives and functionality (DeHaan, 1962).

Socioemotional Approach

The socioemotional approach is where relationship-oriented perceptions and the leader's behavior toward followers are processed. When children process leadership stimuli within this approach, they navigate within a relationship-based dimension guided by personal judgement (Piaget & Gabain, 1932). Mental schemas are nourished by increased awareness of their interpersonal dimensions (Selman et al., 1977), and attention is given to the relationship of the leader with followers, how leaders make decisions, and how they relate to others (Ayman-Nolley & Ayman, 2005; Broich, 1929; DeHaan, 1962; Nemerowicz & Rosi, 1997; Pigors, 1933; Selman & Jaquette, 1977; Yarrow & Campbell, 1963). In this sense, children most often associate leaders with role models from close contexts such as from their family, friendship circle, or school. Cognitive processes turn perceptive attention toward socioemotional processes and relationship-based attributes of leaders (Chauvin & Karnes, 1982; DeHaan, 1962; Pigors, 1933; Selman & Jaquette, 1977) that can heighten critical and sometimes negative views of leaders (Broich, 1929; DeHaan, 1962; Oliveira, 2016).

When the development of children's leadership conceptualizations forms within this dimension, information processing is aligned with self-reflective and reciprocal perspectives as well as third person or mutual perspectives (Selman & Jaquette, 1977), where followership is guided by moral ideas of righteousness

Table 2
Empirical Studies of Children's Conceptualizations of Leadership Not Conducted Within an ILT Perspective

Author	Sample size	Age/grade	Context	Methodology	Main finding
Broich (1929)	58	8–10 years old	Germany	Survey/open answer questionnaire	Younger children more often choose a leader based on physical appearance. The oldest children's representation of a leader is interpreted within the functional leadership system and its context. Girls usually stress character attributes in the leader above the attainment of results. They want their leader to be sympathetic and understanding, as well as good at games, and to be loyal, trustworthy, and easy to get along with. The boys really want the spirit of leadership, (enthusiastic cooperation in view of a common cause) and the results of domination (unyielding discipline, quick, and reliable accomplishment).
Piaget & Gabain, (1932)	20 boys	4–13 years old	Switzerland	Interviews, while children play marbles	Ages 3–8: Egocentric stage, children are exposed and observant to rules by the outside world, focus on the joys of the situation, and are not knowingly interested in innovation or collaboration. Ages 8–10: Cooperation stage, children become interested in the sociality of the game, in the set rules defined by older peers or adults, and enjoy the idea of collaborative play. Ignition of their capacity of judging the acts of others. Ages 10–12: Codification of rules stage, children become interested in the rules; they see them as a code common to all of the social group, with higher awareness of goodness and compassion. Compliance shifts from adult regulation toward personal judgement.
Pigors (1933)	Sample 1: 105 Sample 2: 35	0–10 years old	United States	Interviews	Dominance and leadership are different in how they emerge and develop. Similarities as well as differences between boys and girls in the characteristics they seek in a leader chosen among their own. From an early age, children show interest in leadership demonstrated by the ongoing tendency to copy adult "leaders" such as fathers, mothers, or teachers and incorporate them into their role play.
Gowan (1955)	485	Last 4 years of secondary school	United States (military school)	Questionnaires	Leaders tend to be youinger than the mean of the group, and characteristics such as achievement, ability are significant. Leadership appears to be considered a type of achievement.
White and Lippitt (1960)	Unknown	10–11 years old	United States	Questionnaires applied after a mask-making workshop conducted in one authoritarian and one democratic atmosphere	Indicated that authoritarian leadership fosters group productivity, but it results in less group commitment and more group hostility.
DeHaan (1962)		5–17 years old	United States	Interviews and questionnaires	

(*table continues*)

Table 2 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
	Sample 1: 51 Sample 2: 256 Sample 3: 236				The study provides evidence that children as young as those in kindergarten have a concept of leader linked to observable roles of leaders (the one in front); and as they grow up, their perception moves toward more action-based roles of leaders (telling others what to do); and finally, the concept evolves to a more public-spirited role of leaders (trustworthy, understanding). Children develop well-defined, functional concepts of leaders around the middle elementary grades, sometime around the fourth grade, at the age of 10.
Yarrow and Campbell (1963)	267	8–13 years old	United States (summer camp)	Children's perceptions of one another	Eight-year-old children already have a concept of effective leadership behavior.
Durojaiye (1969)	312	8–11 years old	United States	Sociometric tests	Leadership choices appeared not to be influenced by ethnic background while friendship choices were.
Hardy (1975)	308	Grade 4	United States	Sociometric tests and questionnaires (Least Preferred Coworker scale)	Results support the contingency model. Children in Grade 4 already hold leadership styles.
Selman et al. (1977)	125 children and an unknown number of adults	4–32 years old	United States	Mixed methods including interviews with children and adults, observation, overhearing children and adolescent conversations in school settings and conversations with parents. A part of the study is dedicated to naturalistic observations where researchers asked children at school to "act as a leader" in sports games	Leadership is an issue related to the conception of peer group organization. The leadership styles enacted by a few children in different age groups were consistent with cognitive and social development stages proposed by the authors where younger children have a more physical, practical idea of leadership, in the middle of primary school, they move toward a functional, authoritarian idea of leadership and in the last years, they develop a sense of cooperation.
Selman and Jaquette (1977)	48	7–16 years old	United States	Interview measures on various Piagetian logicophysical problems and interpersonal problems and issues interview	The authors found that children with poor social relations, or from different social classes or gender, may escalate the proposed four stages within the leadership domain at a different pace.
Chauvin and Karnes (1984)	122	10–13 years old Grades 4–7	United States (Summer program for gifted students)	Checklists of Leadership Inventory	Positive self-concept in gifted students. No significant gender differences. Older students give greater significance to the leader being goal-oriented, doing as they say, and being open to feedback.
Daniels-Beirness (1986)	Sample 1: 160 Sample 2: 66	Kindergarten, Grades 2, 4, and 6	Canada	Interviews	Evidence of hierarchical staged developmental model. Older children presented higher levels of reasoning explained by more leadership experience. Possibility of multistage reasoning and shifting trends as children grow older from physical to interpersonal attributes of leadership.
Morris (1991)	58 adolescents 58 adults	Unknown	Canada	Questionnaires	Rank-order between the adolescents and adults is significant.
Edwards (1994)	304	Grades 4–6	United States	Peer-nomination technique	Strong associations between leadership and managerial qualities like organized and goal oriented.
F. A. Karnes (1995)	50	High school juniors and seniors	United States (leadership program)	Unknown	Participants reported strong influence from family members and Jesus Christ as the greatest leader, student council as a key opportunity for leadership, shyness as the biggest obstacle to becoming a leader, and did not give high ratings to the importance of popularity.

(*table continues*)

Table 2 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
Ganz (1996)	100	Grade 8	United States	Written survey, written story responding to cartoon prompt, classroom discussion	Young people defined leadership in terms of characteristics and activities, and rarely in terms of relational aspects. Sport and entertainment contexts have high influence over young people's ideas of leaders. Perceptions of cynicism were associated with leaders. Gender similarities in characteristics noted for good leadership, such as honest and responsible.
F. Karnes et al. (1996)	264	Grades 4–11	United States (gifted students' programs)	Questionnaire (In your opinion, who are the three greatest leaders of this century?)	Point out the student's exposure to a greater diversity of roles is needed. Differences across genders except in the intellectually gifted children who most frequently chose their mother. Boys most often cited historical figures in the leadership group (Martin Luther King Jr.) and girls their mother. In the academically talented group, boys cited most often a historical figure (Adolf Hitler) and girls, most often their teacher.
Nemerowicz and Rosi (1997)	85	Grades 4 and 5 (~8 and ~9 years old)	United States	Drawings and interviews (about drawing and ideas of leadership)	This study found that children associate leadership with a top-down dynamic, referring to authority figures or tasks. Children tend to believe that the leader tells others what to do, is somewhat unreachable, and there is no reference to communal well-being or socioemotional aspects. Nevertheless, they report that children do hope that leaders would be more receptive or aware of others and with a teamwork-based approach. Children emphasized the ideal leader as someone who helps people in need, protects others from harm, and solves problems. Children tend to depict a leader of their own gender, "though boys do so at twice the rate" (p. 29). This study also found gender differences, where boys reported functional characteristics as the most relevant when solving problems. Girls on the other hand, reported relational characteristics as the most relevant. In this age frame, children appear to obtain their idea of leader from politics firstly and secondly from family and school.
B. Schneider et al. (2002)	Ranged from 150 to 174 for the 6-month time lag, from 105 to 125 for the 18-month lag, and from 49 to 92 for the 24-month time lag	Grade 9–12		Self-rated predictor measures: Myers–Briggs Type Indicator, Campbell Interest and Skill Survey, Miner Sentence Completion Scale, group discussions. Leadership behavior ratings by teachers. Peer nominations	Same traits that predict adult leadership predict peer nominations of adolescent leaders. Popularity and leadership are different constructs. Leadership and popularity are more strongly correlated than they are correlated with friendship.
Yamaguchi and Maehr (2004)	249	Grades 4 and 5 (~9 to ~11 years old)	United States	Self-reported surveys adapted from Stogdill's Leadership Behavior Descriptor Questionnaire (Stogdill, 1948)	Relationships between group and classroom characteristics with children's self-perceived emergent leadership, student gender, ability group composition, and classroom context were not related to task- or relationship-focused emergent leadership. Group gender composition seems to influence self-perceived task-focused emergent leadership.

(*table continues*)

Table 2 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
Antonakis and Dalgas (2009)	684 adults 681 children	5–13 years old	Switzerland	Choose from two faces, the captain of their boat	Both adults and children were likely to choose the same candidate in a simulated voting environment.
Salmond and Fleshman (2010)	Sample 1: 165 girls, boys, and mothers Sample 2: 2,475 girls and 1,514 boys Sample 3: 3,284	Sample 1: unknown Sample 2: 8–17 years old Sample 3: 13–17 years old	United States	Focus groups and ethnographies. Survey	Young people associate a leader with authority in terms of power and control. Nevertheless, this definition is the least aspirational. Preferences are those of socioemotional and humanitarian nature.
Mortensen et al. (2014)	130	12–19 years old	United States (leadership program)	Online photovoice	Youth's ideas of leadership are unique. They believe it is available for anyone, with purpose for change, social action, coaching, and mentoring. However, strong character is very relevant.
Nor Amin et al. (2017)[a]	350 from each country	Youth aged between 15 and 24	South Korea and Brunei Darussalam	Questionnaires	Bruneian youth prefer directive leadership while South Korean youth prefer supportive leadership. This finding is based on the selected styles that are desirable by the youth and their choice is also influenced by culture. The study has also discovered various literal constructs using path goal theory, and on the basis of leadership style preferences, a leader can change their leadership style.
Dietl et al. (2018)[a]	819 (460 boys and 359 girls)	Grades 3–6 (~8–~10 years old)	Germany	Children watched an animated story (captain displaying leadership behaviors and accomplishing tasks). Then children provided ratings for six photos of handball players/coaches (neutral and impression-management) depending on which they wanted to replace the captain	Adults' and children's judgments of faces correlate with targets' occupational status and leadership role at their place of work. Core self-evaluations mediated this association, suggesting that having a face that conveys status may contribute to a positive self-concept in men. Consistency between the perceptions of adults and children suggests that the mechanism underlying these relations do not rely on socialization within an organizational framework. Thus, what shows on the "outside" of a person may, in some instances, match a bit of what he or she holds on the "inside," potentially forecasting the efficacy of that individual as a leader.
Stavans and Baillargeon (2019)[a]	120	17-month-old	Unknown	Experiment (infants watched a group of three bear puppets who served as the protagonist, wrongdoer, and victim) and measured look responses and length	Infants in the leader condition looked significantly longer if shown the nonintervention as opposed to the intervention event, suggesting that they expected the leader to intervene and rectify the wrongdoer's transgression. In contrast, infants in the nonleader condition looked equally at the events, suggesting that they held no particular expectation for intervention from the nonleader. By the second year of life, infants thus already ascribe unique responsibilities to leaders, including that of righting wrongs.

(*table continues*)

Table 2 (*continued*)

Author	Sample size	Age/grade	Context	Methodology	Main finding
Scherr et al. (2019)[a]	3,570	6–15 years old	Germany, Argentina, Australia, Canada, China, Denmark, Germany, Iran, Italy, Malaysia, the Netherlands, Slovenia, Spain, and Thailand.	Questionnaires (Likert scale with young interviewers explaining questions on the spot). Measured children's beliefs about parental approval of their emotion expressions, their own approval of characters' emotions, and their self-reports about their emotional displays	The authors considered two cultural emotions (assertive and vulnerable) from the Global Leadership and Organizational Behavior Effectiveness when measuring the assertiveness dimension, asking whether and how the associations among children's emotion displays, parental approval, and their own approval of characters' emotions vary by cultural assertiveness, they also measure humane orientation. They found that associations between perceived parental approval and children's expression of sadness and fear were stronger in countries ranking higher on assertiveness and humane orientation. Associations between children's approval of TV characters' displays of anger and their own self-reported expressions of anger were stronger in countries with higher scores for assertiveness and humane orientation.
Stavans and Diesendruck (2021)[a]	128	5- and 6-year-olds	Israel	Children heard a story involving a hierarchical dyad (a leader and a nonleader) and an egalitarian dyad (two nonleaders), and then assessed protagonists' relative contributions to a collaborative endeavor (Experiments 1 and 2) or relative withdrawals from a common resource pool earned jointly (Experiment 3)	Children expected a leader to contribute more toward a joint goal than its nonleader partner, and to withdraw an equal share (not more) from a common pool. Children thus gave evidence that they construed leaders as more responsible, rather than more entitled, relative to nonleaders.

Note. ILT = implicit leadership theory.
[a] Surfaced in the scoping study.

Table 3
Empirical Studies of Children's Conceptualizations of Leadership

Approach	Response outcome example	Relevant author
1. Physical-spatiotemporal	"Leaders exist"	Broich (1929)
	"The leader is the one in front"	DeHaan (1962)
	"The leader is the grown up"	Selman et al. (1977)
	"The leader smiles"	Matthews et al. (1989)
	"Leaders are good/peaceful"	Daniels-Beirness (1986)
		Lord and Maher (1991)
		Ayman-Nolley and Ayman (2005)
		Stavans and Diesendruck (2021)
2. Functional	"The leader tells me what to do"	Broich (1929)
	"The leader knows what to do"	Pigors (1933)
	"The leader gets the task done"	Stogdill (1948)
	"I can demand and expect the leader to get the task done"	Chauvin and Karnes (1982)
	"The leader achieves our goal"	DeHaan (1962)
	"The leader works with my group to achieve an objective"	Yarrow and Campbell (1963)
	"There are categories of leaders, contextually wise"	Selman and Jaquette (1977)
	"The leader has followers"	Edwards (1994)
	"Leaders can be good or bad at getting the task done"	Nemerowicz and Rosi (1997)
	"I have reservations about the leader"	Ayman-Nolley and Ayman (2005)
	"Leaders can be violent"	L. Liu et al. (2012)
		Oliveira (2016)
3. Socioemotional	"I can judge the leader depending on how they treat and relate to others and make decisions"	Broich (1929)
		Piaget and Gabain (1932)
	"I can recognize that there are leadership styles"	Pigors (1933)
	"I have positive or negative views of leaders"	White and Lippitt (1960)
	"The leader is responsible within my society"	DeHaan (1962)
		Yarrow and Campbell (1963)
		Hardy (1975)
		Selman and Jaquette (1977)
		Chauvin and Karnes (1982)
		Nemerowicz and Rosi (1997)
		Ayman-Nolley and Ayman (2005)
		Salmond and Fleshman (2010)
		L. Liu et al. (2012)
		Oliveira (2016)
4. Humanitarian	"The leader can make a difference in society"	Broich (1929)
	"The leader can stand for humanitarian beliefs"	Piaget and Gabain (1932)
	"The leader achieves common goals and society's well-being"	Pigors (1933)
	"The leader is trustworthy and understanding"	DeHaan (1962)
	"The leader can decide between what they can or should do"	Selman and Jaquette (1977)
	"The leader is open to criticism and feedback"	Chauvin and Karnes (1982)
	"The leader can have immense negative influence over society"	Lord and Maher (1991)
	"I can criticize the leader"	Sacks (2009)
		Mortensen et al. (2014)

and empathy, contrasting spatial, physical, or functional notions (Piaget & Gabain, 1932). Principles of self-reliant followership monitor the dynamics when granting leadership status (Stogdill, 1948). As children abstract themselves away from the interaction with the other, they simultaneously manage their self-perspectives and the other's perspectives (Selman & Jaquette, 1977) recognizing leadership styles (White & Lippitt, 1960).

This socioemotional approach is also seen in broader personality development theory where socioemotional development relates to a child's increasing realization of their capacity to become an active contributor to their social circles and, ultimately, to society (Caspi et al., 2005). In this sense, personality development theory aligns to children ILT development theory as it contributes to our understanding of how children understand the harmonious or dysfunctional nature of the leader. Personality development theory also emphasizes how contextual variation influences advancement and reports across individuals (De Los Reyes et al., 2009). Emotional development is often generated within and managed in the context of social relationships (Scherer et al., 1986). The theory of relations in children's development has been based on the premise that early relationships lay down a preceding system that forms the initiating conditions of development (Sander, 1975) and that further advancements unfold as a derivative advancement upon immersion in wider contexts and interactions (Sroufe, 2000). In this sense, tracking relational exposure to environmental stimuli is critical for the understanding of children's socioemotional perceptions of leadership.

Humanitarian Approach

The humanitarian approach is where children process concerns for promoting human welfare. Within this dimension, the leader schema is intrinsically linked to the leader's capacity to uphold or defend a collective cause and make a difference (Mortensen et al., 2014; Sacks, 2009). Children approach stimuli through a societal perspective (Selman & Jaquette, 1977) considering that the leader's actions are publicly owned, expected to champion change, and stand for humanitarian beliefs and well-being (DeHaan, 1962). Within this mental reasoning, children administer complex forms of communication, applying various levels of significance to the self and the other, both in verbal and nonverbal format, and as part of an intricate social system (Selman et al., 1977). As children recognize the complexity of social systems and a leader's potential for social impact, they focus on prominent and noticeable stimuli underlying the leader's potential for standing for something (DeHaan, 1962); their multifunctional role is set to achieve common goals and society's well-being (Selman & Jaquette, 1977).

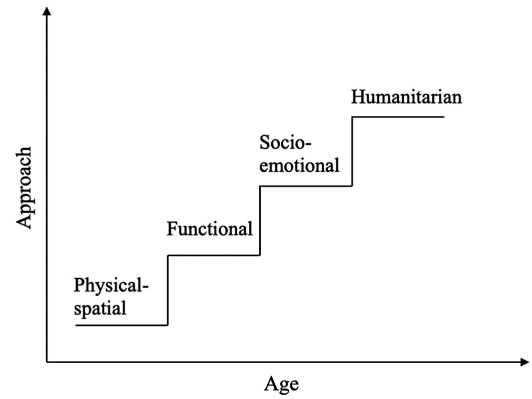
Information is processed within abstract, humanitarian, and idealistic notions (DeHaan, 1962) judged by ideas of goodness and compassion (Piaget & Gabain, 1932). As children navigate across notions of what leadership could do, or should do (Broich, 1929; DeHaan, 1962), they give more importance to the leader/nonleader division (Lord & Maher, 1991) and to stimuli reflecting the public-spirited role of leaders (e.g., trustworthy, understanding; DeHaan, 1962). Consequently, they tend to denote positive community involvement as a key feature in leadership, and nominate as role models, leaders who have stood up for a cause, led change for good, or have caused considerable positive social impact (Mortensen et al., 2014; Sacks, 2009).

Children's ILT development theory is contextually dependent and influenced by their self-concepts (Catrambone et al., 1996; Engle & Lord, 1997; Fong & Markus, 1982; Offermann et al., 1994), which links to other developmental theories such as identity theory that sees development via two lenses; one is externally guided by the influence of social structures over the self (Stryker, 1968), and the other is internally guided by how self-verification affects social behavior (Stets & Burke, 2000). In this sense, identity theory stands at the intersection between the self and society. According to the Wray-Lake and Syvertsen (2011), identity development may precede or overlap with social responsibility development across childhood. For example, they refer to Hart and Fegley's (1995) study where adolescents who showed higher commitment to social action characterized themselves as more ethical and dependable than those with lower commitment to social action. Understanding how children develop humanitarian notions of leaders offers opportunities to cultivate their sense of leadership identity with a focus on prosocial behavior.

Combining the Four Approaches

These four approaches or strategies have traditionally been conceptualized as *sequential,* as illustrated in Figure 2. This logic was naturally underpinned by longstanding ideas about child development (e.g., Piaget & Gabain, 1932). This staircase metaphor approach to children's ILT development suggests that at a certain age and for a prolonged period of time, children think in the same way about a particular topic, then they experience a sudden leap, and reach the next step, where they will think in a new way about the same subject matter. OWT suggests that development across these four approaches is not sequential and may occur arbitrarily or simultaneously at a point in time, although there is a likelihood that some waves are likely to appear before others (see Figure 3). The waves do not compete with each other, but rather reflect different stimuli interacting with individual differences and experiences.

Figure 2
Sequential Development Theory (Piaget) Applied to the Development of Children's ILTs

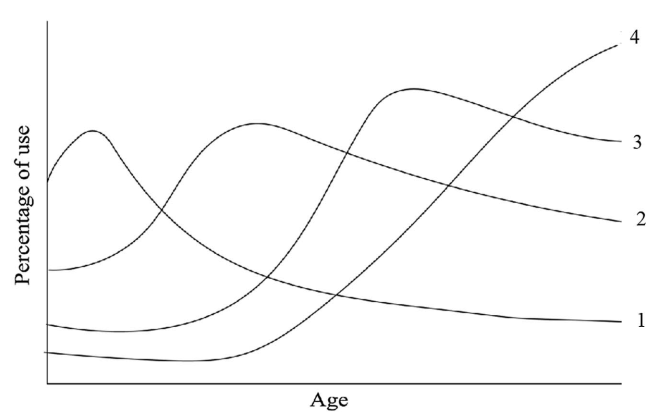


Note. ILT = implicit leadership theory.

A close examination of empirical findings suggests that OWT might better explain the development of children's conceptualizations of leadership than sequential models. The evidence comes from studies showing that children are developing across more than one cognitive dimension simultaneously, the multidimensionality of leadership experiences, differences in escalation, nonage-dependent variation in the manner in which children develop their leadership conceptions, and age differences related to the cognitive dimensions. All of these phenomena make the sequential model a poor fit with the data and are better explained by OWT.

Although previous theories of the emergence of children's leadership conceptualizations have not explored the simultaneous presence of different forms of leadership cognitive development (e.g., Broich, 1929; Chauvin & Karnes, 1982; DeHaan, 1962;

Figure 3
Overlapping Waves Theory Applied to the Development of Children's ILTs



Note. ILT = implicit leadership theory.

Pigors, 1933; Selman et al., 1977; Selman & Jaquette, 1977), some authors recognized multidimensionality and described the possibility of children developing their thinking in more than one way about the concept of leader at the same time. For example, Selman and Jaquette (1977) noted that a child might apply different stages of leadership awareness depending on the social situation, since toward the end of primary/elementary school, children hold functional, relational, and socioemotional ideas of leadership. Selman and Jaquette (1977) also found that by the end of high school, young adults hold a multidimensional understanding of leaders due to their awareness of a leader's potential for social impact. Besides consideration of multidimensionality, Selman et al. (1977) also recognized Piaget's notion of décalage (Paine, 1984; Piaget, 1941; Robson, 2006), where a child draws on the same concept but with increasing depth of understanding or application to different contexts or in different forms at different stages of cognitive development.

Nonage-dependent variation and change in development of social and moral concepts, including that of leadership, has been attributed to children's perception of self and "the other" (Selman et al., 1977), their escalating knowledgeability about their social structure (Ayman-Nolley & Ayman, 2005; Hess & Easton, 1960; Selman et al., 1977), and religious affiliation and education (Oliveira, 2016). Through media such as TV, children expand the contexts they know, either by being more sensitive to such stimuli (Ayman-Nolley & Ayman, 2005; Nemerowicz & Rosi, 1997; Selman et al., 1977) or by being more exposed to them (Ayman-Nolley & Ayman, 2005; Hess & Easton, 1960; Massey, 1975; Okamura, 1968; Oliveira, 2016). Furthermore, the transformation of a positive to a skeptical view of leaders during primary and elementary schooling has also been ascribed to social cognition development where less social awareness may cause more positive views of leaders (Massey, 1975; Okamura, 1968) and, possibly, increased social awareness may cause more idealist or humanitarian views of leadership in young adults (DeHaan, 1962).

In line with Bronfenbrenner's (1979) developmental theory that likens cognitive development to a set of Russian dolls, Selman and Jaquette (1977) proposed that children escalate, at an individual pace, through their understanding of leadership across three domains of interpersonal consciousness: (a) awareness of their individuality, (b) awareness of their relationships with close friends, and (c) awareness of group structures and functionality. This expansion of awareness about their social structure results in children's ongoing increased capacity to recognize more elements of leadership, categories of leaders, leader roles with increased level of detail (Ayman-Nolley & Ayman, 2005; Sacks, 2009), as well as socially recognizable exemplars of leaders, and contextually renowned leader stereotypes such as famous people (Ayman-Nolley & Ayman, 2005; Nemerowicz & Rosi, 1997). In addition, difference in escalation can also be influenced by each child's capacity for social interaction and their appreciation of the social and economic environment (Selman & Jaquette, 1977).

Children's leadership conceptualizations become more sophisticated as they relate to leaders, but also, as they witness or exercise leadership themselves (Sacks, 2009; Salmond & Fleshman, 2010). McClelland and Jenkins (2014), based on Siegler's theories, explain that children apply different procedures to weigh input from experience. More leadership experience brings more knowledge, and in time, this increases the tendency to judge leaders guided by abstract prototypes (Brown & Lord, 2001; Matthews et al., 1989). However, experience, more than age, is the main reason for this progress, since sometimes younger children can hold more sophisticated and abstract understanding of some concepts, than older children (Brown & Lord, 2001; Elman, 2005; McClelland & Jenkins, 2014; Robson, 2006; Seidenberg, 1994).

When exploring children's progression of the understanding of leadership across these dimensions, the literature shows discrepancies in the specific age when children shift from one phase into the other. For example, several studies (Broich, 1929; DeHaan, 1962; Hess & Easton, 1960; Okamura, 1968; Pigors, 1933) found that children do not become knowledgeable of followership until they are 10 years old. On the other hand, Ayman-Nolley and Ayman (2005), Nemerowicz and Rosi (1997), Selman and Jaquette (1977), and Yarrow and Campbell (1963) found this shift earlier, at around 8 years of age, when children can also have socioemotional ideas of leadership regarding the relationship of the leader with followers. Furthermore, Oliveira (2016) proposed that even by 8 years old, children already have moved from a physical notion of leadership into a task-based conception, and that by 10 years old, they already give importance to relationship-based attributes of leaders. Daniels-Beirness' (1986) study of conceptual changes of children's understanding of leadership across age found multistage reasoning and shifting trends as children grow older from physical to interpersonal attributes of leadership. OWT offers a contemporary and useful framework for examining and integrating this literature because it is able to incorporate what have previously been regarded as inconsistencies in the literature. In OWT terms, these stages or orientations can be conceived as cognitive dimensions through which children travel as their conceptualizations of leadership develop.

Discussion

Drawing from published work on the development of children's conceptualizations of leadership and the recent small corps of work looking at the formation of children's ILTs, we applied OWT to better understand how the literature has investigated ILT development during childhood. ILT theory must advance hand in hand with children cognitive developmental science. This application of OWT has provided a useful framework for recent discoveries on children's cognitive development across diverse fields. Applying OWT navigates across transactional theories where genes and environment dynamically interact with each other across time (Bjorklund, 2020), where contextual inferences impact children's learning and influence their development (P. Cantor et al., 2019), and where the vibrant nature of thinking and behavior resolves new approaches as connections and disconnections are constantly recurring (P. Cantor et al., 2019; Robson, 2006; Siegler, 1996).

The application of OWT better reflects children's leadership conceptualization development than the phase or stage-based models previously advocated. According to the OWT, children interpret the world they are immersed in, multiple waves of thinking are stimulated (Barbarin & Wasik, 2009), they decide to which stimuli they will respond, and their choices result in different cognitive development (Baars & Gage, 2010). When children encounter stimuli that is associated with leaders or leadership, multiple cognitive reactions are ignited, and they may choose to process information within "mixed" waters of understanding. This constant creation and expansion of causal connections is what advances knowledge and develops expertise (Robson, 2006; Robson & Flannery Quinn, 2015; Siegler et al., 2003).

Leadership emerges and evolves from complex interactions between individuals through their perceptions of, and reactions to, each other (Bjorklund & Ellis, 2014; Martinko & Gardner, 1987). Ongoing relations shape its meaning; hence, it is individually, collectively, and socially devised (Fairhurst & Grant, 2010). Our application of OWT to the development of children's leadership thinking focuses on the participant's view of the leadership phenomenon. This is negotiated socially and historically (Fairhurst & Grant, 2010) and is seen as a social process that depends on both leaders and followers (Lord & Maher, 1991), where social environments are connected (Z. Liu et al., 2021).

Implications for Future Research

The present review has significantly expanded our understanding of children's ILTs, challenging notions that children's development is broadly age based (Broich, 1929; DeHaan, 1962), or dependent on social cognition (Selman et al., 1977; Selman & Jaquette, 1977), or that children ILTs develop in a stepwise or stage-based manner (DeHaan, 1962; Sacks, 2009; Selman et al., 1977). Through the application of OWT, the present review offers an alternative approach to childhood ILT development aligned with contemporary thinking in the children's cognitive development literature (Bjorklund, 2020; Cohen, 2017; Robson & Flannery Quinn, 2015; Siegler, 1996). It resolves inconsistencies in the literature such as the age when children shift from one phase or stage to the next and asserts that the development of children's leadership conceptualizations is nonlinear, can occur in different dimensions simultaneously, and is largely a response to their experiences. When viewing the current depiction of children's development of ILTs through an OWT lens, the inadequacy of current empirical investigations become more apparent and suggests the need to test various elements, implications, and boundaries of ILTs with methods better suited to capturing dynamic phenomena.

First, as children show a variety of reasoning which unfolds as diverse strategic approaches to solving particular problems (Bjorklund & Causey, 2017; N. Cantor & Mischel, 1979; Siegler, 1996, 2000; Siegler et al., 2003) and leadership conceptualization implies continual changing circumstances across situations and contexts, there is a need for empirical research to study how children navigate across these cognitive dimensions at different times. Such future study suggests longitudinal designs, sampling at short intervals (Siegler, 2007), and capturing within-person change. It should explore where and how each child, in trying to resolve the concept of leader, generates forces of thought that may create high waves, medium waves, overlapping waves, or ripples. This way, we would expand our understanding of the preferred mental pathways when processing leadership stimuli, and what kind of stimuli is generating the energy for the creation of these mental impressions.

Second, a key avenue for future research is to look at what sets transition processes in motion. In terms of the practicalities, these might be the most important elements because they are levers that might be used to influence the development of children's understanding of leadership. In addition to looking at what ignites transition processes, it is important to look at how these waves of learning wax and wane. Is it due to a shortage of experiential stimuli, or perhaps other aspects of leadership assert themselves in a replacement style model, or perhaps some sort of cognitive or learning saturation happens? How do "crucible moments" (A. Byrne et al., 2018, p. 276), both positive and negative, foment or alter the ILT formation process? Answers to these questions move beyond testing the descriptive and explanatory aspects of the application of OWT and focus more on the forces that produce change and how, as Siegler might phrase it, these contribute to the overall tide.

Third, there are research implications for the development of ILTs in adults. Our review focuses on formative development of ILTs throughout childhood. This begs the question of how malleable are ILTs in adulthood (Epitropaki & Martin, 2004). Extraordinary amounts of money are spent every year on leader development (Moldoveanu & Narayandas, 2019), but relatively little is known about the cognitive changes that are expected or achieved. Gabriel (2005) argued that the way adults conceptualize leadership is largely set at a young age and Jackson and Parry (2018) argued that leaders' learning about leadership is formed early in life and changes in adulthood are relatively small. But to what extent is this true? Although adult brains are less plastic than younger ones (Kühn & Lindenberger, 2016), it seems very pessimistic to assume that no cognitive change to ILTs is possible. Exploring this adult domain would help produce a lifelong model of ILT development that would inform the leader development literature and highlight opportunities that exist.

We believe this review will open avenues for future research testing the positivity bias in leadership studies. By highlighting the role of children's successes and failures in shaping the development of their leadership conceptualizations, OWT suggests explorations of positive and negative aspects of ILT dimensions. Future studies could be conducted in contexts outside the school, sport, and summer camps. For example, different perceptions of leaders may be found in youth detention centers. Then, we may begin to discover if the dimension in the model should mutate and be renamed. The positive nomenclature of functional dimension might be challenged when dysfunctional behaviors are observed, such as when the leader is expected to prevent the task to be completed. In the socioemotional context, there might be a choice between socioemotional and socioalexithymia, where the leader might be expected to avoid any kind of emotional interaction with the follower. And in the humanitarian dimension, there could be a bifurcation into humanitarian and maleficent elements depending on the nature of the experience.

We know that 10- and 11-year-old children prefer democratic leadership above autocratic and noninterventionist forms of leadership (White & Lippitt, 1960). Walker et al. (2020) found that adults who lived their adolescence in conflict-infused and achievement-focused family environments showed a higher tendency to factor tyranny as an ideal characteristic in a leader. In contrast, adults who lived adolescence in intellectually stimulating family environments were less inclined to endorse tyranny in their ILTs. These effects on the emergence of children's ILTs are amplified by parents' egalitarian attitudes (Walker et al., 2020). Such results illustrate one process by which the development of children's ILTs might bifurcate to produce the antiprototypes revealed by Epitropaki and Martin (2004) in adult ILTs. Further research is required to explore whether other processes might exist to help us understand how the less desirable ILTs emerge. It will give us insight into how ILTs that are far from humanitarian emerge and develop.

Implications for Practice

It is well-established that children develop their conceptualizations of leadership as they become more socially aware and as they experience or witness leadership (Ayman-Nolley & Ayman, 2005; Broich, 1929; Chauvin & Karnes, 1984; DeHaan, 1962; Hess & Easton, 1960;

Okamura, 1968; Pigors, 1933; Sacks, 2009; Salmond & Fleshman, 2010; Selman et al., 1977). Our review highlights that it is not safe to think that children's understanding of leadership develops in an orderly series of phases and waves may wash over them based on the information they have available (Robson & Flannery Quinn, 2015). This is particularly relevant in an era when there is "dramatically more exposure of children to media … affecting aspects of their development" (Cohen, 2017, p. 127) which is expanding every day. For example, Meta has been planning to launch a version of Instagram for children under 13. Further, this is an age of fake news, misinformation, and disinformation, when trust in the press and politicians is at an all-time low (Hanitzsch et al., 2018) and this is particularly important for leadership, with so much of children's exposure to current affairs and leadership coming through digital third parties. Piecing this together, our application of OWT alerts us to the risks of a tsunami of false and misleading information that could wash over children, disrupting gentler forms of development about the nature of leadership. Hence, the contemporary landscape in which children are developing their ILTs is changing rapidly, challenging the relevance of studies conducted a decade and more ago. Consequently, there are pressing needs for studies of the impact of new technologies and digital media on children's ILTs, opportunities for these advancements to support development (Barbot et al., 2020), how it travels across the ILT cognitive dimensions, and how to leverage these advancements to support development (Barbot et al., 2020).

The application of OWT suggests that there are critical periods of ILT development. Cohen (2017) and Bjorklund (2020) provided evidence that once the brain misses an opportunity to grow, potential connections are lost. For example, physically, in the development of vision, if a baby is not exposed to light, the baby does not develop the neural pathways required for normal vision (Bjorklund, 2020; Cohen, 2017). Cognitively, in the development of language, as noted by Cohen (2017), if children do not hear language by the time they are 7 or 8, they lose the ability to speak normally. This evidence shows that children have critical periods where the brain appears to be receptive which are the foundation for later growth and development. Consequently, certain experiences at a particular time can cause change in the structure and functioning of infant and children's brains (Bjorklund, 2020; Cohen, 2017).

This consideration is of importance for the development of leadership conceptualizations because throughout the learning of a concept, there are moments of threshold concept advancement (Billsberry, 2016; Meyer & Land, 2005; Wright & Gilmore, 2012). This means that, as perceivers advance their acquisition of new knowledge, what is acquired causes perceivers to see the idea or theme in a new way, integrating prior knowledge, demarcating new cognitive boundaries, and transforming their understanding in such a way that it is impossible to forget or reverse (Meyer & Land, 2003). But what happens if perceivers miss the critical moment of development? Do they become stuck in their potential for leadership development and education? So, we need to find out if there are critical moments during childhood where particular stimuli play a powerful role in the formation and development of ILTs.

Limitations

It is well-established that contextual and situational factors influence the way children conceptualize leadership (Ayman-Nolley & Ayman, 2005; Hess & Easton, 1960; Massey, 1975; Okamura, 1968; Sacks, 2009; Selman & Jaquette, 1977; Stogdill, 1948). These include interactions with caregivers and family environments (Massey, 1975; Oliveira, 2016; Pigors, 1933; Rosenblith, 1959; Walters & Stinnett, 1971), perceived parental or primary caregiver traits (Ayman-Nolley & Ayman, 2005; Hunt et al., 1990; Jablin & Kron, 1994; Keller, 1999; L. Liu et al., 2012), school environments and peer group dynamics (Ahlbrand & Reynolds, 1972; Sacks, 2009; Sahgal & Pathak, 2007; Triandis, 2004), religion (Oliveira, 2016), simultaneous immersion in media, books, political settings, and entertainment contexts (Ayman-Nolley & Ayman, 2005; Hess & Easton, 1960; Massey, 1975; Nemerowicz & Rosi, 1997; Okamura, 1968; Sacks, 2009; Stogdill, 1948), policies and programs (Ayman-Nolley & Ayman, 2005), as well as leadership experiences with leaders and leadership (Sacks, 2009; Salmond & Fleshman, 2010). Further, a child's social and economic status and levels of affluence (Salmond & Fleshman, 2010; Selman & Jaquette, 1977) can impact the development of the construct. There are also personal characteristics that have been found to influence the development of ILTs such as children's self-concept (Selman et al., 1977). Also, verbal knowledge and emotional capability (Broich, 1929; DeHaan, 1962), social cognition (Ayman-Nolley & Ayman, 2005; DeHaan, 1962; Hess & Easton, 1960; Selman & Jaquette, 1977), sense of responsibility (Sacks, 2009), and ability, or inability, to cope with their social or/and learning environment (Selman et al., 1977; Selman & Jaquette, 1977). Hence, our application of OWT to the development of leadership conceptualizations during childhood will only cover part of their developmental journey, and we would encourage scholars to capture many of these other factors in their studies along with ILTs.

In addition, extensive research reports that gender impacts children's perceptions of leadership because boys and girls present differences in ideas, preferences, and functional characteristics of leaders (Ayman-Nolley & Ayman, 2005; Ayman-Nolley et al., 2006; Broich, 1929; Eva et al., 2021; Nemerowicz & Rosi, 1997; Selman & Jaquette, 1977; Yamaguchi & Maehr, 2004). Moreover, in early primary/elementary school, children already hold different systems of beliefs, values, and expectations regarding the behavior of women and men (Eccles, 2007; Grusec & Hastings, 2014; Schwartz & Rubel, 2005). Consequently, they present gender-specific leadership role stereotypes (Frost, 2016) and prototypes that influence their leadership identity (Eva et al., 2021).

Last, ethnicity can also influence the development of children's ILTs (Ayman-Nolley & Ayman, 2005; Selman & Jaquette, 1977) and children's leadership behavior (Salmond & Fleshman, 2010). For example, it has been found that while in the 1990s most children in the United States represented male White leaders, in more recent years, African American children more often choose leader prototypes from their own ethnicity (Ayman-Nolley & Ayman, 2005). In addition, there are unique associations to distinctive social roles in studies conducted in United States, China, Costa Rica, Canada, and Philippines; and even within a country with children from different cultural backgrounds (Ayman-Nolley & Ayman, 2005; Ayman-Nolley et al., 2006; L. Liu et al., 2012; Oliveira, 2016).

All these factors, which are often absorbed through role models, are antecedents to leadership conceptualizations (Bandura, 1986; Keller, 1999). Consequently, there are many avenues to develop and enhance the holistic application of OWT, which space constraints do not allow us to do here. We have summarized the research questions emerging from our theoretical development in Table 4.

Table 4
Summary of Research Questions Relating to Children's Conceptualizations of Leadership

Topic	Research questions
Development of ILTs during childhood	1. How do children's mental strategies for processing leadership stimuli appear, change, wax, and wane?
	2. What kind of stimuli generate the creation of these mental impressions?
	3. What are the critical periods of ILT development in childhood?
	4. What are the moments of threshold concept advancement?
	5. Do positive (prototypical) and negative (antiprototypical) elements of ILTs develop in the same way, for the same reasons, and at the same time?
	6. What happens to children's ILTs if they miss key development opportunities?
Influences on ILT development	7. What factors shape the development of children's ILTs?
	8. How do these factors influence the development of their ILTs?
	9. What sets transition processes in motion?
	10. What forces produce change in children's ILTs, how, and when?
	11. What is the impact of new technologies and digital media on children's ILTs?
	12. How does a child's intellectual development interact with the development of his or her ILT?
	13. How does a child's language development interact with the development of his or her ILT?
Behavior and children's ILTs	14. How do children choose preferred strategy(ies) when processing leadership stimuli?
	15. How does strategy choice vary depending on the type of leadership stimuli?
	16. How do children explore positive and negative aspects of ILT dimensions? And how do these develop?
Societal and family influences on ILT development	17. How does gender influence the development of children's ILTs?
	18. How does ethnicity influence the development of children's ILTs?
	19. How does religion influence the development of children's ILTs?
	20. How does family and family background influence the development of children's ILTs?
	21. How does a child's social and economic background influence the development of their ILT?
	22. How does schooling influence the development of children's ILTs?
Relationship between child and adult ILTs	23. How do children's ILTs relate to those of adults?
	24. At what point can a child's ILTs be said to be their adult ILT?
	25. How does the change in ILTs that happens during development relate to change in ILTs that happens once fully formed?
	26. Are the factors influencing the malleability of ILTs in adulthood similar to those occurring during childhood?

Note. ILT = implicit leadership theory.

Conclusion

This article makes an important contribution to the current body of knowledge in ILTs by reviewing and reinterpreting the findings of empirical studies into children's conceptualizations of leadership according to OWT. This approach explains these findings better than age-based or stepwise, phase-based theories, which struggle to capture the multidimensionality of children's leadership thinking. Instead, OWT explains nonage-related variation, and accounts for nonlinear progression that can occur in different cognitive dimensions simultaneously, and is largely a response to their experiences. This approach allows for the study of change, which is crucial when exploring strategies and factors that influence transformation in leadership thinking across the early years.

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